AgarwalSingle

Pearson correlation, r

Cell type	Original performance	MPRALegnet	Mprann	Malinois	PARM	DREAM-RNN	Modern-gena base*	ModernGENA large*	Caduceus PS*	DNA-BERT2*	NTv3 650M*
HepG2	**0,8189**	0,806	0,7736	0.7274	0.7985	0.7767	0.6925 ± 0.0032	0.7294 ± 0.0021	0.7216 ± 0.0115	0.6787 ± 0.0025	0.7257 ± 0.0146
K562	**0,8514**	0,83	0,7901	0.7816	0.823	0.8012	0.7559 ± 0.0045	0.7783 ± 0.0031	0.7649 ± 0.0047	0.7141 ± 0.0077	0.7804 ± 0.0059
WTC11	**0,7354**	0,718	0,6796	0.6254	0.7237	0.6724	0.7162 ± 0.0039	0.7415 ± 0.0027	0.6509 ± 0.0125	0.6715 ± 0.0065	0.7078 ± 0.0086

K-fold on Modern-gena base, ModernGENA large, Caduceus PS, DNA-BERT2, NTv3 650M.

AgarwalMulti

Pearson correlation, r

Cell type	Original performance	MPRALegnet	Mprann	Malinois	PARM	DREAM_RNN	Modern-gena base*	ModernGENA large*	Caduceus PS*	DNA-BERT2*	NTv3 650M*
HepG2	0,78	**0,798**	0,7633	0.708170	0.7886	0.742	0.6947 ± 0.0078	0.7392 ± 0.0148	0.7136 ± 0.0086	0.6923 ± 0.0062	0.7191 ± 0.0095
K562	0,75	**0,759**	0,7248	0.666216	0.755	0.6883	0.6506 ± 0.0066	0.7056 ± 0.0127	0.6615 ± 0.0170	0.6421 ± 0.0065	0.6779 ± 0.0068
WTC11	0,77	**0,77**	0,738	0.690842	0.7658	0.7034	0.6972 ± 0.0086	0.7344 ± 0.0138	0.6876 ± 0.0123	0.6727 ± 0.0052	0.7073 ± 0.0041

K-fold on Modern-gena base, ModernGENA large, Caduceus PS, DNA-BERT2, NTv3 650M.

Gosai2024

Pearson correlation, r

Cell type	Original performance	MPRALegnet	Mprann	Malinois	PARM	DREAM-RNN	Modern-gena base*	ModernGENA large*	Caduceus PS*	DNA-BERT2*	NTv3 650M*
HepG2	0,89	**0,908**	0,8727	0,8704	0,8991	0,8971	0.8611	0.8729	0.8687	0.7962	0.8905
K562	0,88	**0,906**	0,8648	0,8663	0,8969	0,8909	0.8561	0.8668	0.8621	0.7931	0.8844
SK-N-SH	0,88	**0,898**	0,867	0,8631	0,8894	0,888	0.8585	0.8672	0.8588	0.7969	0.8801

deAlmeida2022

Cel type	Original performance	MPRALegnet	Mprann	Malinois	PARM	DREAM-RNN	Modern-gena base*	ModernGENA large*	Caduceus PS*	DNA-BERT2*	NTv3 650M*
Developmental	0,68	0.7274	0,7121	0.6871	0.7117	0.7295	0.6672	0.6901	0.6415	0.6797	0.6944
Housekeeping	0,74	0.7962	0,7815	0.7751	0.7845	0.7986	0.7635	0.7838	0.7560	0.7694	0.7856